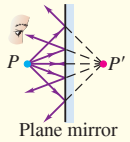
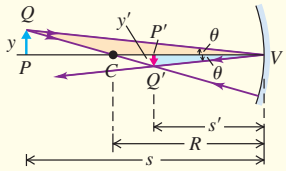
# CHAPTER 34 SUMMARY

**Reflection or refraction at a plane surface:** When rays diverge from an object point $P$ and are reflected or refracted, the directions of the outgoing rays are the same as though they had diverged from a point $P'$ called the image point. If they actually converge at $P'$ and diverge again beyond it, $P'$ is a real image of $P$; if they only appear to have diverged from $P'$, it is a virtual image. Images can be either erect or inverted.

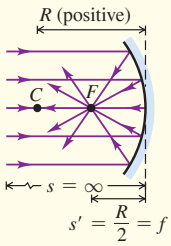



**Lateral magnification:** The lateral magnification $m$ in any reflecting or refracting situation is defined as the ratio of image height $y'$ to object height $y$. When $m$ is positive, the image is erect; when $m$ is negative, the image is inverted.

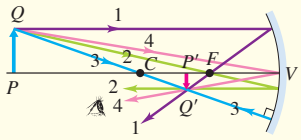
$$m = \frac{y'}{y} \qquad (34.2)$$




**Focal point and focal length:** The focal point of a mirror is the point where parallel rays converge after reflection from a concave mirror, or the point from which they appear to diverge after reflection from a convex mirror. Rays diverging from the focal point of a concave mirror are parallel after reflection; rays converging toward the focal point of a convex mirror are parallel after reflection. The distance from the focal point to the vertex is called the focal length, denoted as $f$. The focal points of a lens are defined similarly.




**Relating object and image distances:** The formulas for object distance $s$ and image distance $s'$ for plane and spherical mirrors and single refracting surfaces are summarized in the table. The equation for a plane surface can be obtained from the corresponding equation for a spherical surface by setting $R = \infty$. (See Examples 34.1–34.7.)



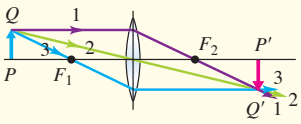

| | Plane Mirror | Spherical Mirror | Plane Refracting Surface | Spherical Refracting Surface |
|---|---|---|---|---|
| Object and image distances | $\frac{1}{s} + \frac{1}{s'} = 0$ | $\frac{1}{s} + \frac{1}{s'} = \frac{2}{R} = \frac{1}{f}$ | $\frac{n_a}{s} + \frac{n_b}{s'} = 0$ | $\frac{n_a}{s} + \frac{n_b}{s'} = \frac{n_b - n_a}{R}$ |
| Lateral magnification | $m = -\frac{s'}{s} = 1$ | $m = -\frac{s'}{s}$ | $m = -\frac{n_a s'}{n_b s} = 1$ | $m = -\frac{n_a s'}{n_b s}$ |

Object–image relationships derived in this chapter are valid for only rays close to and nearly parallel to the optic axis; these are called paraxial rays. Nonparaxial rays do not converge precisely to an image point. This effect is called spherical aberration.

**Thin lenses:** The object–image relationship, given by Eq. (34.16), is the same for a thin lens as for a spherical mirror. Equation (34.19), the lensmaker's equation, relates the focal length of a lens to its index of refraction and the radii of curvature of its surfaces. (See Examples 34.8–34.11.)

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f} \qquad (34.16)$$

$$\frac{1}{f} = (n - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \qquad (34.19)$$

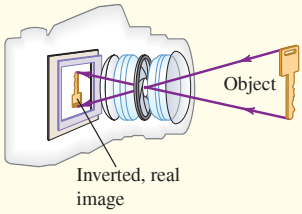



**Sign rules:** The following sign rules are used with all plane and spherical reflecting and refracting surfaces:

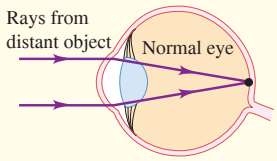
- $s > 0$ when the object is on the incoming side of the surface (a real object); $s < 0$ otherwise.
- $s' > 0$ when the image is on the outgoing side of the surface (a real image); $s' < 0$ otherwise.
- $R > 0$ when the center of curvature is on the outgoing side of the surface; $R < 0$ otherwise.
- $m > 0$ when the image is erect; $m < 0$ when inverted.

**Cameras:** A camera forms a real, inverted image of the object being photographed on a light-sensitive surface. The amount of light striking this surface is controlled by the shutter speed and the aperture. The intensity of this light is inversely proportional to the square of the $f$-number of the lens. (See Example 34.12.)

$$f\text{-number} = \frac{\text{Focal length}}{\text{Aperture diameter}} = \frac{f}{D} \qquad (34.20)$$

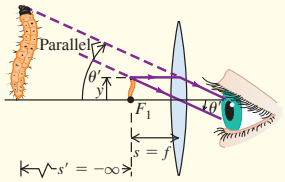



**The eye:** In the eye, refraction at the surface of the cornea forms a real image on the retina. Adjustment for various object distances is made by squeezing the lens, thereby making it bulge and decreasing its focal length. A nearsighted eye is too long for its lens; a farsighted eye is too short. The power of a corrective lens, in diopters, is the reciprocal of the focal length in meters. (See Examples 34.13 and 34.14.)

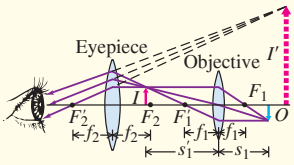



**The simple magnifier:** The simple magnifier creates a virtual image whose angular size $\theta'$ is larger than the angular size $\theta$ of the object itself at a distance of 25 cm, the nominal closest distance for comfortable viewing. The angular magnification $M$ of a simple magnifier is the ratio of the angular size of the virtual image to that of the object at this distance.

$$M = \frac{\theta'}{\theta} = \frac{25\text{ cm}}{f} \qquad (34.22)$$




**Microscopes and telescopes:** In a compound microscope, the objective lens forms a first image in the barrel of the instrument, and the eyepiece forms a final virtual image, often at infinity, of the first image. The telescope operates on the same principle, but the object is far away. In a reflecting telescope, the objective lens is replaced by a concave mirror, which eliminates chromatic aberrations.




Chapter 34 Media Assets

For assigned homework and other learning materials, go to Mastering Physics.

## GUIDED PRACTICE

### KEY EXAMPLE VARIATION PROBLEMS

**Be sure to review EXAMPLES 34.1, 34.3, and 34.4 (Section 34.2) before attempting these problems.**

**VP34.4.1** You place a light bulb 15.0 cm in front of a concave spherical mirror. The mirror forms an image of the bulb on a wall 4.50 m in front of the mirror. Find (a) the mirror's focal length, (b) the mirror's radius of curvature, and (c) the lateral magnification of the image.

**VP34.4.2** A concave spherical mirror has radius of curvature 37.0 cm. Find the image distance and lateral magnification for each of the following object distances. In each case state whether the image is real or virtual, whether it is erect or inverted, and whether it is larger or smaller than the object. (a) 11.0 cm; (b) 31.0 cm; (c) 55.0 cm.

**VP34.4.3** A spherical mirror has radius of curvature −44.0 cm. When you look at your eye in the mirror, your eye's reflection appears to be 18.0 cm behind the mirror's surface. (a) Is the mirror concave or convex? (b) How far is your eye from the mirror? (c) What is the lateral magnification of the image of your eye? Is the image real or virtual? Erect or inverted? Larger or smaller than your eye?

**VP34.4.4** A convex spherical mirror has radius of curvature −37.0 cm. Find the image distance and lateral magnification for each of the following object distances. In each case state whether the image is real or virtual, whether it is erect or inverted, and whether it is larger or smaller than the object. (a) 11.0 cm; (b) 31.0 cm; (c) 55.0 cm.

**Be sure to review EXAMPLE 34.8 (Section 34.4) before attempting these problems.**

**VP34.8.1** Both sides of a double convex thin lens have radii of curvature of the same magnitude. The lens is made of glass with index of refraction 1.65, and the focal length of the lens is +30.0 cm. Find the radius of curvature of (a) the front surface and (b) the back surface.

**VP34.8.2** The side of a thin lens that faces the object is convex and has radius of curvature 15.0 cm. The other side is concave and has radius of curvature 25.0 cm. The lens is made of glass with index of refraction 1.55. (a) Is this lens thicker at its center or at its edges? (b) What is the focal length of the lens? (c) Is the lens converging or diverging?

**VP34.8.3** The side of a thin lens that faces the object is convex and has radius of curvature 25.0 cm. The other side is concave and has radius of curvature 15.0 cm. The lens is made of glass with index of refraction 1.55. (a) Is this lens thicker at its center or at its edges? (b) What is the focal length of the lens? (c) Is the lens converging or diverging?

**VP34.8.4** You are designing a lens to be made of glass with index of refraction 1.70. The first surface (the surface toward the object) is to be convex with radius of curvature 28.0 cm, and the focal length of the lens is to be 14.0 cm. (a) What must be the radius of curvature of the second surface (the surface away from the object)? (b) Will the second surface be concave or convex?

**Be sure to review EXAMPLES 34.9 and 34.10 (Section 34.4) before attempting these problems.**

**VP34.10.1** You place a strawberry 15.0 cm in front of a converging lens with focal length +25.0 cm. (a) What is the image position? Is it on the same side of the lens as the strawberry or on the opposite side? (b) What is the lateral magnification of the image? (c) Is the image real or virtual? Erect or inverted? Larger or smaller than the strawberry?

**VP34.10.2** If you put an eraser 28.0 cm in front of a lens, an image of the eraser is formed 42.0 cm behind the lens. (a) Find the focal length of the lens. Is the lens converging or diverging? (b) What is the lateral magnification of the image? (c) Is the image real or virtual? Erect or inverted? Larger or smaller than the eraser?

**VP34.10.3** When you place a drawing pin 48.0 cm in front of a lens, the resulting image is real, inverted, and the same size as the drawing pin. (a) Is the lens converging or diverging? (b) What is the image distance? (c) What is the focal length of the lens?

**VP34.10.4** The front door of many houses has a small "peephole" lens that gives a wide-angle view of the outside. To provide this wide-angle view, the lens must produce an image of an object outside that is erect and smaller than the object. For one lens, if an object 20.0 cm tall is 2.50 m in front of the door, its erect image is only 4.00 cm tall. (a) What is the lateral magnification of the image? (b) What is the image distance? Is the image on the same side of the lens as the object or on the opposite side? (c) What is the focal length of the lens? Is the lens converging or diverging?

## BRIDGING PROBLEM Image Formation by a Wine Goblet

A thick-walled wine goblet can be considered to be a hollow glass sphere with an outer radius of 4.00 cm and an inner radius of 3.40 cm. The index of refraction of the goblet glass is 1.50. (a) A beam of parallel light rays enters the side of the empty goblet along a horizontal radius. Where, if anywhere, will an image be formed? (b) The goblet is filled with white wine ($n = 1.37$). Where is the image formed?

**SOLUTION GUIDE**

### IDENTIFY and SET UP

1. The goblet is *not* a thin lens, so you cannot use the thin-lens formula. Instead, you must think of the inner and outer surfaces of the goblet walls as spherical refracting surfaces. The image formed by one surface serves as the object for the next surface. Draw a diagram that shows the goblet and the light rays that enter it.
2. Choose the appropriate equation that relates the image and object distances for a spherical refracting surface.

### EXECUTE

3. For the empty goblet, each refracting surface has glass on one side and air on the other. Find the position of the image formed by the first surface, the outer wall of the goblet. Use this as the object for the second surface (the inner wall of the same side of the goblet) and find the position of the second image. (*Hint:* Be sure to account for the thickness of the goblet wall.)
4. Continue the process of step 3. Consider the refractions at the inner and outer surfaces of the glass on the opposite side of the goblet, and find the position of the final image. (*Hint:* Be sure to account for the distance between the two sides of the goblet.)
5. Repeat steps 3 and 4 for the case in which the goblet is filled with wine.

### EVALUATE

6. Are the images real or virtual? How can you tell?

# PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

## DISCUSSION QUESTIONS

**Q34.1** A spherical mirror is cut in half horizontally. Will an image be formed by the bottom half of the mirror? If so, where will the image be formed?

**Q34.2** For the situation shown in Fig. 34.3, is the image distance $s'$ positive or negative? Is the image real or virtual? Explain your answers.

**Q34.3** The laws of optics also apply to electromagnetic waves invisible to the eye. A satellite TV dish is used to detect radio waves coming from orbiting satellites. Why is a curved reflecting surface (a "dish") used? The dish is always concave, never convex; why? The actual radio receiver is placed on an arm and suspended in front of the dish. How far in front of the dish should it be placed?

**Q34.4** Explain why the focal length of a *plane* mirror is infinite, and explain what it means for the focal point to be at infinity.

**Q34.5** If a spherical mirror is immersed in water, does its focal length change? Explain.

**Q34.6** For what range of object positions does a concave spherical mirror form a real image? What about a convex spherical mirror?

**Q34.7** When a room has mirrors on two opposite walls, an infinite series of reflections can be seen. Discuss this phenomenon in terms of images. Why do the distant images appear fainter?

**Q34.8** For a spherical mirror, if $s = f$, then $s' = \infty$, and the lateral magnification $m$ is infinite. Does this make sense? If so, what does it mean?

**Q34.9** You may have noticed a small convex mirror next to your bank's ATM. Why is this mirror convex, as opposed to flat or concave? What considerations determine its radius of curvature?

**Q34.10** A student claims that she can start a fire on a sunny day using just the sun's rays and a concave mirror. How is this done? Is the concept of image relevant? Can she do the same thing with a convex mirror? Explain.

**Q34.11** A person looks at his reflection in the concave side of a shiny spoon. Is it right side up or inverted? Does it matter how far his face is from the spoon? What if he looks in the convex side? (Try this yourself!)

**Q34.12** In Example 34.4 (Section 34.2), there appears to be an ambiguity for the case $s = 10\,\text{cm}$ as to whether $s'$ is $+\infty$ or $-\infty$ and whether the image is erect or inverted. How is this resolved? Or is it?

**Q34.13** Suppose that in the situation of Example 34.7 of Section 34.3 (see Fig. 34.26) a vertical arrow 2.00 m tall is painted on the side of the pool beneath the water line. According to the calculations in the example, this arrow would appear to the person shown in Fig. 34.26 to be 1.50 m long. But the discussion following Eq. (34.13) states that the magnification for a plane refracting surface is $m = 1$, which suggests that the arrow would appear to the person to be 2.00 m long. How can you resolve this apparent contradiction?

**Q34.14** The bottom of the passenger-side mirror on your car notes, "Objects in mirror are closer than they appear." Is this true? Why?

**Q34.15** How could you very quickly make an approximate measurement of the focal length of a converging lens? Could the same method be applied if you wished to use a diverging lens? Explain.

**Q34.16** The focal length of a simple lens depends on the color (wavelength) of light passing through it. Why? Is it possible for a lens to have a positive focal length for some colors and negative for others? Explain.

**Q34.17** When a converging lens is immersed in water, does its focal length increase or decrease in comparison with the value in air? Explain.

**Q34.18** A spherical air bubble in water can function as a lens. Is it a converging or diverging lens? How is its focal length related to its radius?

**Q34.19** Can an image formed by one reflecting or refracting surface serve as an object for a second reflection or refraction? Does it matter whether the first image is real or virtual? Explain.

**Q34.20** If a piece of photographic film is placed at the location of a real image, the film will record the image. Can this be done with a virtual image? How might one record a virtual image?

**Q34.21** According to the discussion in Section 34.2, light rays are reversible. Are the formulas in the table in this chapter's Summary still valid if object and image are interchanged? What does reversibility imply with respect to the *forms* of the various formulas?

**Q34.22** You've entered a survival contest that will include building a crude telescope. You are given a large box of lenses. Which two lenses do you pick? How do you quickly identify them?

**Q34.23** **BIO** You can't see clearly underwater with the naked eye, but you *can* if you wear a face mask or goggles (with air between your eyes and the mask or goggles). Why is there a difference? Could you instead wear eyeglasses (with water between your eyes and the eyeglasses) in order to see underwater? If so, should the lenses be converging or diverging? Explain.

**Q34.24** You take a lens and mask it so that light can pass through only the bottom half of the lens. How does the image formed by the masked lens compare to the image formed before masking?

## EXERCISES

### Section 34.1 Reflection and Refraction at a Plane Surface

**34.1** • A candle 4.85 cm tall is 39.2 cm to the left of a plane mirror. Where is the image formed by the mirror, and what is the height of this image?

**34.2** • The image of a tree just covers the length of a plane mirror 3.70 cm tall when the mirror is held 34.0 cm from the eye. The tree is 20.0 m from the mirror. What is its height?

**34.3** • A pencil that is 9.0 cm long is held perpendicular to the surface of a plane mirror with the tip of the pencil lead 12.0 cm from the mirror surface and the end of the eraser 21.0 cm from the mirror surface. What is the length of the image of the pencil that is formed by the mirror? Which end of the image is closer to the mirror surface: the tip of the lead or the end of the eraser?

### Section 34.2 Reflection at a Spherical Surface

**34.4** • A concave mirror has a radius of curvature of 36.0 cm. (a) What is its focal length? (b) If the mirror is immersed in water (refractive index 1.33), what is its focal length?

**34.5** • An object 0.600 cm tall is placed 16.5 cm to the left of the vertex of a concave spherical mirror having a radius of curvature of 22.0 cm. (a) Draw a principal-ray diagram showing the formation of the image. (b) Determine the position, size, orientation, and nature (real or virtual) of the image.

**34.6** • Repeat Exercise 34.5 for the case in which the mirror is convex.

**34.7** •• The diameter of Mars is 6794 km, and its minimum distance from the earth is $5.58 \times 10^7$ km. When Mars is at this distance, find the diameter of the image of Mars formed by a spherical, concave telescope mirror with a focal length of 1.85 m.

**34.8** •• An object is 24.1 cm from the center of a spherical silvered-glass Christmas tree ornament 5.80 cm in diameter. What are the position and magnification of its image?

**34.9** • A coin is placed next to the convex side of a thin spherical glass shell having a radius of curvature of 17.0 cm. Reflection from the surface of the shell forms an image of the 1.5-cm-tall coin that is 6.50 cm behind the glass shell. Where is the coin located? Determine the size, orientation, and nature (real or virtual) of the image.

**34.10** • You hold a spherical salad bowl 85 cm in front of your face with the bottom of the bowl facing you. The bowl is made of polished metal with a 30 cm radius of curvature. (a) Where is the image of your 2.0-cm-tall nose located? (b) What are the image's size, orientation, and nature (real or virtual)?

**34.11** • A spherical, concave shaving mirror has a radius of curvature of 32.1 cm. (a) What is the magnification of a person's face when it is 11.2 cm to the left of the vertex of the mirror? (b) Where is the image? Is the image real or virtual? (c) Draw a principal-ray diagram showing the formation of the image.

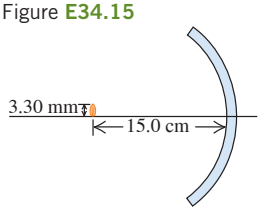
**34.12** • For a concave spherical mirror that has focal length $f = +18.0$ cm, what is the distance of an object from the mirror's vertex if the image is real and has the same height as the object?

**34.13** • **Dental Mirror.** A dentist uses a curved mirror to view teeth on the upper side of the mouth. Suppose she wants an erect image with a magnification of 2.00 when the mirror is 1.25 cm from a tooth. (Treat this problem as though the object and image lie along a straight line.) (a) What kind of mirror (concave or convex) is needed? Use a ray diagram to decide, without performing any calculations. (b) What must be the focal length and radius of curvature of this mirror? (c) Draw a principal-ray diagram to check your answer in part (b).

**34.14** • For a convex spherical mirror that has focal length $f = -12.0$ cm, what is the distance of an object from the mirror's vertex if the height of the image is half the height of the object?

**34.15** • The thin glass shell shown in **Fig. E34.15** has a spherical shape with a radius of curvature of 12.0 cm, and both of its surfaces can act as mirrors. A seed 3.30 mm high is placed 15.0 cm from the center of the mirror along the optic axis, as shown in the figure. (a) Calculate the location and height of the image of this seed. (b) Suppose now that the shell is reversed. Find the location and height of the seed's image.

Figure E34.15




**34.16** • An object 0.600 cm tall is placed 24.0 cm to the left of the vertex of a concave spherical mirror. The image of the object is inverted and is 2.50 cm tall. What is the radius of curvature of the mirror?

**34.17** • A shiny spoon provides both a concave mirror and a convex mirror, one on each side. (a) Hold a shiny spoon about 25 cm from your face. Observe your image in the concave side. Is your image up-side down or right-side up? (b) Does that mean that your image is real or virtual? (c) Estimate the height of your head. (d) Estimate the height of the image of your head. (Use the size of the spoon as a guide.) (e) Using these estimates, determine the magnification. (f) Use Eqs. (34.6) and (34.5) to estimate the radius of curvature of the spoon. (g) Now observe your image in the convex side of the spoon. Is it up-side down or right-side up? (h) Is your image real or virtual?

## Section 34.3 Refraction at a Spherical Surface

**34.18** •• A tank whose bottom is a mirror is filled with water to a depth of 20.2 cm. A small fish floats motionless 7.5 cm under the surface of the water. (a) What is the apparent depth of the fish when viewed at normal incidence? (b) What is the apparent depth of the image of the fish when viewed at normal incidence?

**34.19** • A speck of dirt is embedded 3.90 cm below the surface of a sheet of ice ($n = 1.309$). What is its apparent depth when viewed at normal incidence?

**34.20** • Parallel rays from a distant object are traveling in air and then are incident on the concave end of a glass rod with a radius of curvature of 15.0 cm. The refractive index of the glass is 1.50. What is the distance between the vertex of the glass surface and the image formed by the refraction at the concave surface of the rod? Is the image in the air or in the glass?

**34.21** • A person swimming 1.00 m below the surface of the water in a swimming pool looks at the diving board that is directly overhead and sees the image of the board that is formed by refraction at the surface of the water. This image is a height of 4.00 m above the swimmer. What is the actual height of the diving board above the surface of the water?

**34.22** • A person is lying on a diving board 3.00 m above the surface of the water in a swimming pool. She looks at a coin that is on the bottom of the pool directly below her. To her, the coin appears to be a distance of 7.00 m from her. What is the depth of the water at this point?

**34.23** •• **A Spherical Fish Bowl.** A small tropical fish is at the center of a water-filled, spherical fish bowl 28.0 cm in diameter. (a) Find the apparent position and magnification of the fish to an observer outside the bowl. The effect of the thin walls of the bowl may be ignored. (b) A friend advised the owner of the bowl to keep it out of direct sunlight to avoid blinding the fish, which might swim into the focal point of the parallel rays from the sun. Is the focal point actually within the bowl?

**34.24** • The left end of a long glass rod 6.00 cm in diameter has a convex hemispherical surface 3.00 cm in radius. The refractive index of the glass is 1.60. Determine the position of the image if an object is placed in air on the axis of the rod at the following distances to the left of the vertex of the curved end: (a) infinitely far; (b) 12.0 cm; (c) 2.00 cm.

**34.25** •• The glass rod of Exercise 34.24 is immersed in oil ($n = 1.45$). An object placed to the left of the rod on the rod's axis is to be imaged 1.20 m inside the rod. How far from the left end of the rod must the object be located to form the image?

**34.26** •• The left end of a long glass rod 8.00 cm in diameter, with an index of refraction of 1.60, is ground and polished to a convex hemispherical surface with a radius of 4.00 cm. An object in the form of an arrow 1.50 mm tall, at right angles to the axis of the rod, is located on the axis 24.0 cm to the left of the vertex of the convex surface. Find the position and height of the image of the arrow formed by paraxial rays incident on the convex surface. Is the image erect or inverted?

**34.27** • A 3.20-mm-tall object is 20.0 cm from the vertex of a spherical mirror. The mirror forms an image 60.0 cm from the mirror. (a) If the image is real, what is the radius of curvature of the mirror? What is the height of the image? Is it upright or inverted? (b) Repeat part (a) for the case where the image is virtual.

## Section 34.4 Thin Lenses

**34.28** •• To determine the focal length $f$ of a converging thin lens, you place a 4.00-mm-tall object a distance $s$ to the left of the lens and measure the height $h'$ of the real image that is formed to the right of the lens. You repeat this process for several values of $s$ that produce a real image. After graphing your results as $1/h'$ versus $s$, both in cm, you find that they lie close to a straight line that has slope 0.208 $\text{cm}^{-2}$. What is the focal length of the lens?

**34.29** • An insect 3.75 mm tall is placed 25.0 cm to the left of a thin planoconvex lens. The left surface of this lens is flat, the right surface has a radius of curvature of magnitude 12.9 cm, and the index of refraction of the lens material is 1.70. (a) Calculate the location and size of the image this lens forms of the insect. Is it real or virtual? Erect or inverted? (b) Repeat part (a) if the lens is reversed.

**34.30** • A lens forms an image of an object. The object is 16.0 cm from the lens. The image is 12.0 cm from the lens on the same side as the object. (a) What is the focal length of the lens? Is the lens converging or diverging? (b) If the object is 8.50 mm tall, how tall is the image? Is it erect or inverted? (c) Draw a principal-ray diagram.

**34.31** • A converging meniscus lens (see Fig. 34.32a) with a refractive index of 1.52 has spherical surfaces whose radii are 7.00 cm and 4.00 cm. What is the position of the image if an object is placed 24.0 cm to the left of the lens? What is the magnification?

**34.32** • A converging lens with a focal length of 70.0 cm forms an image of a 3.20-cm-tall real object that is to the left of the lens. The image is 4.50 cm tall and inverted. Where are the object and image located in relation to the lens? Is the image real or virtual?

**34.33** •• A converging lens forms an image of a 7.50-mm-tall real object. The image is 12.5 cm to the left of the lens, 3.30 cm tall, and erect. What is the focal length of the lens? Where is the object located?

**34.34** • A photographic slide is to the left of a lens. The lens projects an image of the slide onto a wall 6.10 m to the right of the slide. The image is 75.0 times the size of the slide. (a) How far is the slide from the lens? (b) Is the image erect or inverted? (c) What is the focal length of the lens? (d) Is the lens converging or diverging?

**34.35** •• A double-convex thin lens has surfaces with equal radii of curvature of magnitude 3.00 cm. Using this lens, you observe that it forms an image of a very distant tree at a distance of 1.89 cm from the lens. What is the index of refraction of the lens?

**34.36** • A converging lens with a focal length of 7.00 cm forms an image of a 3.20-mm-tall real object that is to the left of the lens. The image is 1.70 cm tall and erect. Where are the object and image located? Is the image real or virtual?

**34.37** • A thin lens is made of glass that has refractive index $n = 1.50$. The lens is surrounded by air. The left-hand surface of the lens is flat and the right-hand spherical surface is convex with radius of curvature 20.0 cm, so the lens is thicker in the middle than at its edges. What is the height of the image formed by the lens for a 6.00-mm-tall object that is placed 20.0 cm to the left of the center of the lens?

**34.38** •• A lensmaker wants to make a magnifying glass from glass that has an index of refraction $n = 1.55$ and a focal length of 20.0 cm. If the two surfaces of the lens are to have equal radii, what should that radius be?

**34.39** •• A thin lens is made of glass that has refractive index $n = 1.50$. The lens is surrounded by air. The left-hand spherical surface of the lens is concave with radius of curvature $R$ and the right-hand side is flat, so the lens is thinner in the middle than at its edges. An object is placed 12.0 cm to the left of the center of the lens. What is the value of $R$ if the image formed by the lens is 8.00 cm to the left of the center of the lens?

**34.40** • A converging lens with a focal length of 12.0 cm forms a virtual image 8.00 mm tall, 17.0 cm to the right of the lens. Determine the position and size of the object. Is the image erect or inverted? Are the object and image on the same side or opposite sides of the lens? Draw a principal-ray diagram for this situation.

**34.41** • Repeat Exercise 34.40 for the case in which the lens is diverging, with a focal length of −48.0 cm.

**34.42** • An object is 16.0 cm to the left of a lens. The lens forms an image 36.0 cm to the right of the lens. (a) What is the focal length of the lens? Is the lens converging or diverging? (b) If the object is 8.00 mm tall, how tall is the image? Is it erect or inverted? (c) Draw a principal-ray diagram.

**34.43** •• **Combination of Lenses I.** A 1.20-cm-tall object is 50.0 cm to the left of a converging lens of focal length 40.0 cm. A second converging lens, this one having a focal length of 60.0 cm, is located 300.0 cm to the right of the first lens along the same optic axis. (a) Find the location and height of the image (call it $I_1$) formed by the lens with a focal length of 40.0 cm. (b) $I_1$ is now the object for the second lens. Find the location and height of the image produced by the second lens. This is the final image produced by the combination of lenses.

**34.44** •• **Combination of Lenses II.** Repeat Exercise 34.43 using the same lenses except for the following changes: (a) The second lens is a *diverging* lens having a focal length of magnitude 60.0 cm. (b) The first lens is a *diverging* lens having a focal length of magnitude 40.0 cm. (c) Both lenses are diverging lenses having focal lengths of the same *magnitudes* as in Exercise 34.43.

**34.45** •• **Combination of Lenses III.** Two thin lenses with a focal length of magnitude 12.0 cm, the first diverging and the second converging, are located 9.00 cm apart. An object 2.50 mm tall is placed 20.0 cm to the left of the first (diverging) lens. (a) How far from this first lens is the final image formed? (b) Is the final image real or virtual? (c) What is the height of the final image? Is it erect or inverted? (*Hint:* See the preceding two exercises.)

**34.46** • The focal points of a thin diverging lens are 25.0 cm from the center of the lens. An object is placed to the left of the lens, and the lens forms an image of the object that is 18.0 cm from the lens. (a) Is the image to the left or right of the lens? (b) How far is the object from the center of the lens? (c) Is the height of the image less than, greater than, or the same as the height of the object?

**34.47** • Figure 34.37e shows the principal-ray diagram of the image formation by a converging lens with focal length $f$ and object distance $s = 2f/3$ to the left of the lens. (a) Use Eq. (34.17) to find the image distance $s'$ in terms of $f$. Based on your result, is the image real or virtual? Is it to the left or right of the lens? (b) If the height of the object is $h$, use Eq. (34.17) to find the height of the image. Does that mean the image is upright or inverted? (c) Do your results agree with what is shown in Fig. 34.37e?

**34.48** •• An object is to the left of a thin lens. The lens forms an image on a screen that is 2.60 m to the right of the object. The height of the image is 2.50 times the height of the object. (a) Is the image upright or inverted? (b) What is the focal length of the lens? Is the lens converging or diverging?

## Section 34.5 Cameras

**34.49** • When a camera is focused, the lens is moved away from or toward the digital image sensor. If you take a picture of your friend, who is standing 3.90 m from the lens, using a camera with a lens with an 85 mm focal length, how far from the sensor is the lens? Will the whole image of your friend, who is 175 cm tall, fit on a sensor that is 24 mm × 36 mm?

**34.50** • You wish to project the image of a slide on a screen 9.00 m from the lens of a slide projector. (a) If the slide is placed 15.0 cm from the lens, what focal length lens is required? (b) If the dimensions of the picture on a 35 mm color slide are 24 mm × 36 mm, what is the minimum size of the projector screen required to accommodate the image?

**34.51** • **Zoom Lens.** Consider the simple model of the zoom lens shown in Fig. 34.43a. The converging lens has focal length $f_1 = 12$ cm, and the diverging lens has focal length $f_2 = -12$ cm. The lenses are separated by 4 cm as shown in Fig. 34.43a. (a) For a distant object, where is the image of the converging lens? (b) The image of the converging lens serves as the object for the diverging lens. What is the object distance for the diverging lens? (c) Where is the final image? Compare your answer to Fig. 34.43a. (d) Repeat parts (a), (b), and (c) for the situation shown in Fig. 34.43b, in which the lenses are separated by 8 cm.

## Section 34.6 The Eye

**34.52** •• BIO **Curvature of the Cornea.** In a simplified model of the human eye, the aqueous and vitreous humors and the lens all have a refractive index of 1.40, and all the bending occurs at the cornea, whose vertex is 2.60 cm from the retina. What should be the radius of curvature of the cornea such that the image of an object 40.0 cm from the cornea's vertex is focused on the retina?

**34.53** •• BIO (a) Where is the near point of an eye for which a contact lens with a power of +2.95 diopters is prescribed? (b) Where is the far point of an eye for which a contact lens with a power of −1.60 diopters is prescribed for distant vision?

**34.54** • BIO **Contact Lenses.** Contact lenses are placed right on the eyeball, so the distance from the eye to an object (or image) is the same as the distance from the lens to that object (or image). A certain person can see distant objects well, but his near point is 45.0 cm from his eyes instead of the usual 25.0 cm. (a) Is this person nearsighted or farsighted? (b) What type of lens (converging or diverging) is needed to correct his vision? (c) If the correcting lenses will be contact lenses, what focal length lens is needed and what is its power in diopters?

**34.55** •• BIO **Ordinary Glasses.** Ordinary glasses are worn in front of the eye and usually 2.0 cm in front of the eyeball. Suppose that the person in Exercise 34.54 prefers ordinary glasses to contact lenses. What focal length lenses are needed to correct his vision, and what is their power in diopters?

**34.56** • BIO A person can see clearly up close but cannot focus on objects beyond 75.0 cm. She opts for contact lenses to correct her vision. (a) Is she nearsighted or farsighted? (b) What type of lens (converging or diverging) is needed to correct her vision? (c) What focal length contact lens is needed, and what is its power in diopters?

**34.57** •• A woman wears bifocal glasses with the lenses 2.0 cm in front of her eyes. The upper half of each lens has power −0.500 diopter and corrects her far vision so that she can focus clearly on distant objects when looking through that half. The lower half of each lens has power +2.00 diopters and corrects her near vision when she looks through that half. (a) What are the far point and near point of her eyes? (b) While the woman is repairing a leaky pipe under her kitchen sink, she looks at close objects through the upper half of her bifocal lenses. What is the closest object that she can see clearly?

## Section 34.7 The Magnifier

**34.58** •• A thin lens with a focal length of 6.00 cm is used as a simple magnifier. (a) What angular magnification is obtainable with the lens if the object is at the focal point? (b) When an object is examined through the lens, how close can it be brought to the lens? Assume that the image viewed by the eye is at the near point, 25.0 cm from the eye, and that the lens is very close to the eye.

**34.59** • The focal length of a simple magnifier is 6.00 cm. Assume the magnifier is a thin lens placed very close to the eye. (a) How far in front of the magnifier should an object be placed if the image is formed at the observer's near point, 25.0 cm in front of her eye? (b) If the object is 4.50 mm high, what is the height of its image formed by the magnifier?

**34.60** • You want to view through a magnifier an insect that is 2.10 mm long. If the insect is to be at the focal point of the magnifier, what focal length will give the image of the insect an angular size of 0.026 radian?

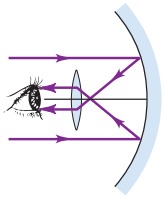
### Section 34.8 Microscopes and Telescopes

**34.61** •• A telescope is constructed from two lenses with focal lengths of 95.0 cm and 15.0 cm, the 95.0 cm lens being used as the objective. Both the object being viewed and the final image are at infinity. (a) Find the angular magnification for the telescope. (b) Find the height of the image formed by the objective of a building 60.0 m tall, 3.00 km away. (c) What is the angular size of the final image as viewed by an eye very close to the eyepiece?

**34.62** •• The eyepiece of a refracting telescope (see Fig. 34.53) has a focal length of 9.00 cm. The distance between objective and eyepiece is 1.20 m, and the final image is at infinity. What is the angular magnification of the telescope?

**34.63** •• A reflecting telescope (**Fig. E34.63**) is to be made by using a spherical mirror with a radius of curvature of 1.28 m and an eyepiece with a focal length of 1.60 cm. The final image is at infinity. (a) What should the distance between the eyepiece and the mirror vertex be if the object is taken to be at infinity? (b) What will the angular magnification be?

Figure E34.63



**34.64** • A compound microscope has an objective lens with focal length 14.0 mm and an eyepiece with focal length 20.0 mm. The final image is at infinity. The object to be viewed is placed 2.0 mm beyond the focal point of the objective lens. (a) What is the distance between the two lenses? (b) Without making the approximation $s_1 \approx f_1$, use $M = m_1 M_2$ with $m_1 = -s_1'/s_1$ to find the overall angular magnification of the microscope. (c) What is the percentage difference between your result and the result obtained if the approximation $s_1 \approx f_1$ is used to find $M$?

**34.65** •• The overall angular magnification of a microscope is $M = -178$. The eyepiece has focal length 15.0 mm and the final image is at infinity. The separation between the two lenses is 202 mm. What is the focal length of the objective? Do not use the approximation $s_1 \approx f_1$ in the expression for $M$.

## PROBLEMS

**34.66** • Where must you place an object in front of a concave mirror with radius $R$ so that the image is erect and $2\frac{1}{2}$ times the size of the object? Where is the image?

**34.67** •• A concave mirror is to form an image of the filament of a headlight lamp on a screen 5.00 m from the mirror. The filament is 4.50 mm tall, and the image is to be 23.0 cm tall. (a) How far in front of the vertex of the mirror should the filament be placed? (b) What should be the radius of curvature of the mirror?

**34.68** • A light bulb is 3.00 m from a wall. You are to use a concave mirror to project an image of the bulb on the wall, with the image 3.50 times the size of the object. How far should the mirror be from the wall? What should its radius of curvature be?

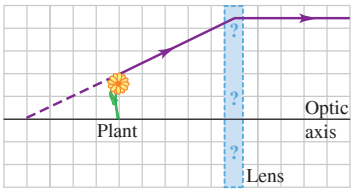
**34.69** •• **CP CALC** You are in your car driving on a highway at 26 m/s when you glance in the passenger-side mirror (a convex mirror with radius of curvature 150 cm) and notice a truck approaching. If the image of the truck is approaching the vertex of the mirror at a speed of 2.5 m/s when the truck is 2.0 m from the mirror, what is the speed of the truck relative to the highway?

**34.70** •• A layer of benzene ($n = 1.50$) that is 4.40 cm deep floats on water ($n = 1.33$) that is 6.00 cm deep. What is the apparent distance from the upper benzene surface to the bottom of the water when you view these layers at normal incidence?

**34.71** •• **Rear-View Mirror.** A mirror on the passenger side of your car is convex and has a radius of curvature with magnitude 18.0 cm. (a) Another car is behind your car, 9.00 m from the mirror, and this car is viewed in the mirror by your passenger. If this car is 1.5 m tall, what is the height of the image? (b) The mirror has a warning attached that objects viewed in it are closer than they appear. Why is this so?

**34.72** •• **Figure P34.72** shows a small plant near a thin lens. The ray shown is one of the principal rays for the lens. Each square is 2.0 cm along the horizontal direction, but the vertical direction is not to the same scale. Use information from the diagram for the following: (a) Using only the ray shown, decide what type of lens (converging or diverging) this is. (b) What is the focal length of the lens? (c) Locate the image by drawing the other two principal rays. (d) Calculate where the image should be, and compare this result with the graphical solution in part (c).

Figure P34.72




**34.73** •• **Pinhole Camera.** A pinhole camera is just a rectangular box with a tiny hole in one face. The film is on the face opposite this hole, and that is where the image is formed. The camera forms an image *without* a lens. (a) Make a clear ray diagram to show how a pinhole camera can form an image on the film without using a lens. (*Hint:* Put an object outside the hole, and then draw rays passing through the hole to the opposite side of the box.) (b) A certain pinhole camera is a box that is 20 cm square and 22.5 cm deep, with the hole in the middle of one of the 20 cm × 20 cm faces. If this camera is used to photograph a fierce chicken that is 20 cm high and 1.4 m in front of the camera, how large is the image of this bird on the film? What is the lateral magnification of this camera?

**34.74** •• An object with height 4.00 mm is placed 28.0 cm to the left of a converging lens that has focal length 8.40 cm. A second lens is placed 8.00 cm to the right of the converging lens. (a) What is the focal length of the second lens if the final image is inverted relative to the 4.00-mm-tall object and has height 5.60 mm? (b) What is the distance between the original object and the final image?

**34.75** •• What should be the index of refraction of a transparent sphere in order for paraxial rays from an infinitely distant object to be brought to a focus at the vertex of the surface opposite the point of incidence?

**34.76** •• Fill a transparent cylindrical drinking glass with water. Hold a small coin between two fingers submerged in the water immediately inside the glass, and then view the coin from outside the glass. Watch the coin as you move it steadily toward the back of the glass. (a) Does the coin appear to widen as you move it backward in the glass? (b) Estimate the apparent horizontal magnification $m_{\text{back}}$ when the coin is flush with the back side of the glass. (c) Since the curvature is in a horizontal plane, this horizontal magnification is the same as the lateral magnification analyzed in Section 34.3. Use Eqs. (34.11) and (34.12) to derive an expression for the horizontal magnification $m$ as a function of the object position $s$, the magnitude of the radius of curvature of the glass $|R|$, and the index of refraction $n$ of the water. (d) Use your result to derive an expression for $n$ as a function of $m_{\text{back}}$. (e) Substitute your estimate for $m_{\text{back}}$ into your equation for $n$ to estimate the index of refraction of the water.

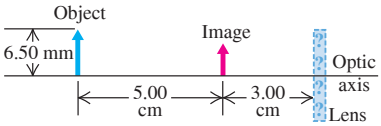
**34.77** •• (a) You want to use a lens with a focal length of 35.0 cm to produce a real image of an object, with the height of the image twice the height of the object. What kind of lens do you need, and where should the object be placed? (b) Suppose you want a virtual image of the same object, with the same magnification—what kind of lens do you need, and where should the object be placed?

**34.78** •• **Autocollimation.** You place an object alongside a white screen, and a plane mirror is 60.0 cm to the right of the object and the screen, with the surface of the mirror tilted slightly from the perpendicular to the line from object to mirror. You then place a converging lens between the object and the mirror. Light from the object passes through the lens, reflects from the mirror, and passes back through the lens; the image is projected onto the screen. You adjust the distance of the lens from the object until a sharp image of the object is focused on the screen. The lens is then 22.0 cm from the object. Because the screen is alongside the object, the distance from object to lens is the same as the distance from screen to lens. (a) Draw a sketch that shows the locations of the object, lens, plane mirror, and screen. (b) What is the focal length of the lens?

**34.79** •• A lens forms a real image that is 214 cm away from the object and $1\frac{2}{3}$ times its height. What kind of lens is this, and what is its focal length?

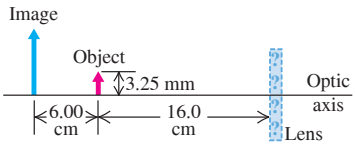
**34.80** • **Figure P34.80** shows an object and its image formed by a thin lens. (a) What is the focal length of the lens, and what type of lens (converging or diverging) is it? (b) What is the height of the image? Is it real or virtual?

Figure P34.80




**34.81** • **Figure P34.81** shows an object and its image formed by a thin lens. (a) What is the focal length of the lens, and what type of lens (converging or diverging) is it? (b) What is the height of the image? Is it real or virtual?

Figure P34.81



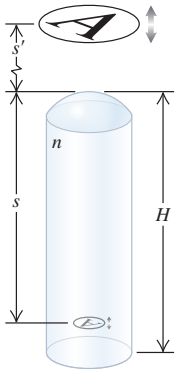

**34.82** ••• A transparent rod 30.0 cm long is cut flat at one end and rounded to a hemispherical surface of radius 10.0 cm at the other end. A small object is embedded within the rod along its axis and halfway between its ends, 15.0 cm from the flat end and 15.0 cm from the vertex of the curved end. When the rod is viewed from its flat end, the apparent depth of the object is 8.20 cm from the flat end. What is its apparent depth when the rod is viewed from its curved end?

**34.83** • BIO **Focus of the Eye.** The cornea of the eye has a radius of curvature of approximately 0.50 cm, and the aqueous humor behind it has an index of refraction of 1.35. The thickness of the cornea itself is small enough that we shall neglect it. The depth of a typical human eye is around 25 mm. (a) What would have to be the radius of curvature of the cornea so that it alone would focus the image of a distant mountain on the retina, which is at the back of the eye opposite the cornea? (b) If the cornea focused the mountain correctly on the retina as described in part (a), would it also focus the text from a computer screen on the retina if that screen were 25 cm in front of the eye? If not, where would it focus that text: in front of or behind the retina? (c) Given that the cornea has a radius of curvature of about 5.0 mm, where does it actually focus the mountain? Is this in front of or behind the retina? Does this help you see why the eye needs help from a lens to complete the task of focusing?

**34.84** • The radii of curvature of the surfaces of a thin converging meniscus lens are $R_1 = +12.0$ cm and $R_2 = +28.0$ cm. The index of refraction is 1.60. (a) Compute the position and size of the image of an object in the form of an arrow 5.00 mm tall, perpendicular to the lens axis, 45.0 cm to the left of the lens. (b) A second converging lens with the same focal length is placed 3.15 m to the right of the first. Find the position and size of the final image. Is the final image erect or inverted with respect to the original object? (c) Repeat part (b) except with the second lens 45.0 cm to the right of the first.

**34.85** ••• CP CALC A transparent cylindrical tube with radius $r = 1.50$ cm has a flat circular bottom and a top that is convex as seen from above, with radius of curvature of magnitude 2.50 cm. The cylinder is filled with quinoline, a colorless highly refractive liquid with index of refraction $n = 1.627$. Near the bottom of the tube, immersed in the liquid, is a luminescent LED display mounted on a platform whose height may be varied. The display is the letter A inside a circle that has a diameter of 1.00 cm. A real image of this display is formed at a height $s'$ above the top of the tube, as shown in **Fig. P34.85**. (a) What is the minimum tube height $H$ for which this display apparatus can function? (b) The luminescent object is moved up and down periodically so that the real image moves up and down in the air above the tube. A mist in the air renders this display visible and dramatic. If we want the image to move from 50.0 cm above the top of the cylinder to 1.00 m above the top of the cylinder during this motion, what is the corresponding range of motion for the object distance $s$? (c) What is the height $s'$ of the image when the object is halfway through its motion? (d) The object in part (b) is moved sinusoidally according to $s = A\sin(\omega t)$, where $A$ is the amplitude and $\omega$ is the angular frequency. The frequency of the motion is 1.00 Hz. What is the velocity of the image when the object is at its midpoint and traveling upward? (e) What is the diameter of the image when it is at its largest size?

Figure P34.85



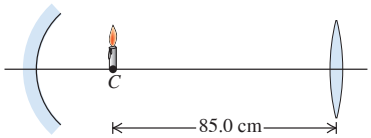

**34.86** • An object is placed 22.0 cm from a screen. (a) At what two points between object and screen may a converging lens with a 3.00 cm focal length be placed to obtain an image on the screen? (b) What is the magnification of the image for each position of the lens?

**34.87** •• BIO A person with a near point of 85 cm, but excellent distant vision, normally wears corrective glasses. But he loses them while traveling. Fortunately, he has his old pair as a spare. (a) If the lenses of the old pair have a power of +2.25 diopters, what is his near point (measured from his eye) when he is wearing the old glasses if they rest 2.0 cm in front of his eye? (b) What would his near point be if his old glasses were contact lenses with the same power instead?

**34.88** •• A screen is placed a distance $d$ to the right of an object. A converging lens with focal length $f$ is placed between the object and the screen. In terms of $f$, what is the smallest value $d$ can have for an image to be in focus on the screen?

**34.89** •• As shown in **Fig. P34.89** (next page), the candle is at the center of curvature of the concave mirror, whose focal length is 10.0 cm. The converging lens has a focal length of 32.0 cm and is 85.0 cm to the right of the candle. The candle is viewed looking through the lens from the right. The lens forms two images of the candle. The first is formed by light passing directly through the lens. The second image is formed from the light that goes from the candle to the mirror, is reflected, and then passes through the lens. (a) For each image, draw a principal-ray diagram that locates the image. (b) For each image, answer the following questions: (i) Where is the image? (ii) Is the image real or virtual? (iii) Is the image erect or inverted with respect to the original object?

Figure P34.89




**34.90** •• **Two Lenses in Contact.** (a) Prove that when two thin lenses with focal lengths $f_1$ and $f_2$ are placed *in contact*, the focal length of the combination is given by the relationship

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

(b) A converging meniscus lens (see Fig. 34.32a) has an index of refraction of 1.55 and radii of curvature for its surfaces of magnitudes 4.50 cm and 9.00 cm. The concave surface is placed upward and filled with carbon tetrachloride ($CCl_4$), which has $n = 1.46$. What is the focal length of the $CCl_4$–glass combination?

**34.91** ••• When an object is placed at the proper distance to the left of a converging lens, the image is focused on a screen 30.0 cm to the right of the lens. A diverging lens is now placed 15.0 cm to the right of the converging lens, and it is found that the screen must be moved 19.2 cm farther to the right to obtain a sharp image. What is the focal length of the diverging lens?

**34.92** •• (a) Repeat the derivation of Eq. (34.19) for the case in which the lens is totally immersed in a liquid of refractive index $n_{liq}$. (b) A lens is made of glass that has refractive index 1.60. In air, the lens has focal length +18.0 cm. What is the focal length of this lens if it is totally immersed in a liquid that has refractive index 1.42?

**34.93** ••• A convex spherical mirror with a focal length of magnitude 24.0 cm is placed 20.0 cm to the left of a plane mirror. An object 0.250 cm tall is placed midway between the surface of the plane mirror and the vertex of the spherical mirror. The spherical mirror forms multiple images of the object. Where are the two images of the object formed by the spherical mirror that are closest to the spherical mirror, and how tall is each image?

**34.94** •• BIO **What Is the Smallest Thing We Can See?** The smallest object we can resolve with our eye is limited by the size of the light receptor cells in the retina. In order for us to distinguish any detail in an object, its image cannot be any smaller than a single retinal cell. Although the size depends on the type of cell (rod or cone), a diameter of a few microns ($\mu$m) is typical near the center of the eye. We shall model the eye as a sphere 2.50 cm in diameter with a single thin lens at the front and the retina at the rear, with light receptor cells 5.0 $\mu$m in diameter. (a) What is the smallest object you can resolve at a near point of 25 cm? (b) What angle is subtended by this object at the eye? Express your answer in units of minutes ($1° = 60$ min), and compare it with the typical experimental value of about 1.0 min. (*Note:* There are other limitations, such as the bending of light as it passes through the pupil, but we shall ignore them here.)

**34.95** • Three thin lenses, each with a focal length of 40.0 cm, are aligned on a common axis; adjacent lenses are separated by 52.0 cm. Find the position of the image of a small object on the axis, 80.0 cm to the left of the first lens.

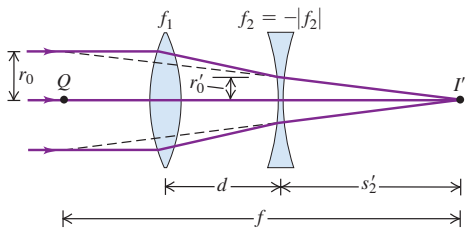
**34.96** •• A camera with a 90-mm-focal-length lens is focused on an object 1.30 m from the lens. To refocus on an object 6.50 m from the lens, by how much must the distance between the lens and the sensor be changed? To refocus on the more distant object, is the lens moved toward or away from the sensor?

**34.97** •• BIO In one form of cataract surgery the person's natural lens, which has become cloudy, is replaced by an artificial lens. The refracting properties of the replacement lens can be chosen so that the person's eye focuses on distant objects. But there is no accommodation, and glasses or contact lenses are needed for close vision. What is the power, in diopters, of the corrective contact lenses that will enable a person who has had such surgery to focus on the page of a book at a distance of 24 cm?

**34.98** •• BIO **A Nearsighted Eye.** A certain very nearsighted person cannot focus on anything farther than 36.0 cm from the eye. Consider the simplified model of the eye described in Exercise 34.52. If the radius of curvature of the cornea is 0.75 cm when the eye is focusing on an object 36.0 cm from the cornea vertex and the indexes of refraction are as described in Exercise 34.52, what is the distance from the cornea vertex to the retina? What does this tell you about the shape of the nearsighted eye?

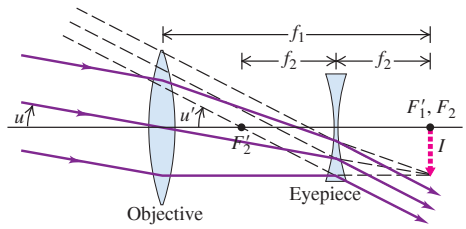
**34.99** ••• **Focal Length of a Zoom Lens. Figure P34.99** shows a simple version of a zoom lens. The converging lens has focal length $f_1$ and the diverging lens has focal length $f_2 = -|f_2|$. The two lenses are separated by a variable distance $d$ that is always less than $f_1$. Also, the magnitude of the focal length of the diverging lens satisfies the inequality $|f_2| > (f_1 - d)$. To determine the effective focal length of the combination lens, consider a bundle of parallel rays of radius $r_0$ entering the converging lens. (a) Show that the radius of the ray bundle decreases to $r_0' = r_0(f_1 - d)/f_1$ at the point that it enters the diverging lens. (b) Show that the final image $I'$ is formed a distance $s_2' = |f_2|(f_1 - d)/(|f_2| - f_1 + d)$ to the right of the diverging lens. (c) If the rays that emerge from the diverging lens and reach the final image point are extended backward to the left of the diverging lens, they will eventually expand to the original radius $r_0$ at some point $Q$. The distance from the final image $I'$ to the point $Q$ is the *effective focal length* $f$ of the lens combination; if the combination were replaced by a single lens of focal length $f$ placed at $Q$, parallel rays would still be brought to a focus at $I'$. Show that the effective focal length is given by $f = f_1|f_2|/(|f_2| - f_1 + d)$. (d) If $f_1 = 12.0$ cm, $f_2 = -18.0$ cm, and the separation $d$ is adjustable between 0 and 4.0 cm, find the maximum and minimum focal lengths of the combination. What value of $d$ gives $f = 30.0$ cm?

Figure P34.99




**34.100** •• **The Galilean Telescope. Figure P34.100** is a diagram of a *Galilean telescope,* or *opera glass,* with both the object and its final image at infinity. The image $I$ serves as a virtual object for the eyepiece. The final image is virtual and erect. (a) Prove that the angular magnification is $M = -f_1/f_2$. (b) A Galilean telescope is to be constructed with the same objective lens as in Exercise 34.61. What focal length should the eyepiece have if this telescope is to have the same magnitude of angular magnification as the one in Exercise 34.61? (c) Compare the lengths of the telescopes.

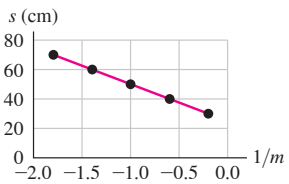
Figure P34.100



**34.101** •• **DATA** It is your first day at work as a summer intern at an optics company. Your supervisor hands you a diverging lens and asks you to measure its focal length. You know that with a *converging* lens, you can measure the focal length by placing an object a distance $s$ to the left of the lens, far enough from the lens for the image to be real, and viewing the image on a screen that is to the right of the lens. By adjusting the position of the screen until the image is in sharp focus, you can determine the image distance $s'$ and then use Eq. (34.16) to calculate the focal length $f$ of the lens. But this procedure won't work with a diverging lens—by itself, a diverging lens produces only virtual images, which can't be projected onto a screen. Therefore, to determine the focal length of a diverging lens, you do the following: First you take a *converging* lens and measure that, for an object 20.0 cm to the left of the lens, the image is 29.7 cm to the right of the lens. You then place a *diverging* lens 20.0 cm to the right of the converging lens and measure the final image to be 42.8 cm to the right of the converging lens. Suspecting some inaccuracy in measurement, you repeat the lens-combination measurement with the same object distance for the converging lens but with the diverging lens 25.0 cm to the right of the converging lens. You measure the final image to be 31.6 cm to the right of the converging lens. (a) Use both lens-combination measurements to calculate the focal length of the diverging lens. Take as your best experimental value for the focal length the average of the two values. (b) Which position of the diverging lens, 20.0 cm to the right or 25.0 cm to the right of the converging lens, gives the tallest image?

**34.102** •• **DATA** In setting up an experiment for a high school biology lab, you use a concave spherical mirror to produce real images of a 4.00-mm-tall firefly. The firefly is to the right of the mirror, on the mirror's optic axis, and serves as a real object for the mirror. You want to determine how far the object must be from the mirror's vertex (that is, object distance $s$) to produce an image of a specified height. First you place a square of white cardboard to the right of the object and find what its distance from the vertex needs to be so that the image is sharply focused on it. Next you measure the height of the sharply focused images for five values of $s$. For each $s$ value, you then calculate the lateral magnification $m$. You find that if you graph your data with $s$ on the vertical axis and $1/m$ on the horizontal axis, then your measured points fall close to a straight line. (a) Explain why the data plotted this way should fall close to a straight line. (b) Use the graph in **Fig. P34.102** to calculate the focal length of the mirror. (c) How far from the mirror's vertex should you place the object in order for the image to be real, 8.00 mm tall, and inverted? (d) According to Fig. P34.102, starting from the position that you calculated in part (c), should you move the object closer to the mirror or farther from it to increase the height of the inverted, real image? What distance should you move the object in order to increase the image height from 8.00 mm to 12.00 mm? (e) Explain why $1/m$ approaches zero as $s$ approaches 25 cm. Can you produce a sharp image on the cardboard when $s = 25$ cm? (f) Explain why you can't see sharp images on the cardboard when $s < 25$ cm (and $m$ is positive).

Figure **P34.102**




**34.103** •• **DATA** The science museum where you work is constructing a new display. You are given a glass rod that is surrounded by air and was ground on its left-hand end to form a hemispherical surface there. You must determine the radius of curvature of that surface and the index of refraction of the glass. Remembering the optics portion of your physics course, you place a small object to the left of the rod, on the rod's optic axis, at a distance $s$ from the vertex of the hemispherical surface. You measure the distance $s'$ of the image from the vertex of the surface, with the image being to the right of the vertex. Your measurements are as follows:

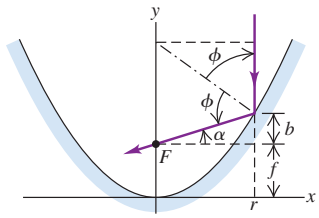
| $s$ (cm) | 22.5 | 25.0 | 30.0 | 35.0 | 40.0 | 45.0 |
|---|---|---|---|---|---|---|
| $s'$ (cm) | 271.6 | 148.3 | 89.4 | 71.1 | 60.8 | 53.2 |

Recalling that the object–image relationships for thin lenses and spherical mirrors include reciprocals of distances, you plot your data as $1/s'$ versus $1/s$. (a) Explain why your data points plotted this way lie close to a straight line. (b) Use the slope and $y$-intercept of the best-fit straight line to your data to calculate the index of refraction of the glass and the radius of curvature of the hemispherical surface of the rod. (c) Where is the image if the object distance is 15.0 cm?

## CHALLENGE PROBLEMS

**34.104** ••• In a spherical mirror, rays parallel to but relatively distant from the optic axis do not reflect precisely to the focal point, and this causes spherical aberration of images. In parabolic mirrors, by contrast, all paraxial rays that enter the mirror, arbitrarily distant from the optic axis, converge to a focal point on the optic axis without any approximation whatsoever. We can prove this and determine the location of the focal point by considering **Fig. P34.104**. The mirror is a parabola defined by $y = ax^2$, with $a$ in units of $(\text{distance})^{-1}$, rotated around the $y$-axis. Consider the ray that enters the mirror a distance $r$ from the axis. (a) What is the slope of the parabola at the point where the ray strikes the mirror, in terms of $a$ and $r$? (b) What is the slope of the dashed line that is normal to the curve at that point? (c) What is the angle $\phi$? (d) Using trigonometry, find the angle $\alpha$ in terms of $\phi$. (e) Using trigonometry, find the distance $b$ in terms of $a$ and $r$. [*Hint:* The identities $\tan(\theta - \pi/2) = -\cot\theta$ and $\cot(2\theta) = (\cot^2\theta - 1)/2\cot\theta$ may be useful.] (f) Find the distance $f$ in terms of $a$. Note that your answer does not depend on the distance $r$; this proves our assertion.

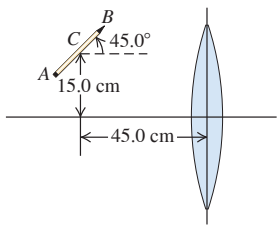
Figure **P34.104**




**34.105** ••• **CALC** (a) For a lens with focal length $f$, find the smallest distance possible between the object and its real image. (b) Graph the distance between the object and the real image as a function of the distance of the object from the lens. Does your graph agree with the result you found in part (a)?

**34.106** ••• **An Object at an Angle.** A 16.0-cm-long pencil is placed at a 45.0° angle, with its center 15.0 cm above the optic axis and 45.0 cm from a lens with a 20.0 cm focal length as shown in **Fig. P34.106**. (Note that the figure is not drawn to scale.) Assume that the diameter of the lens is large enough for the paraxial approximation to be valid. (a) Where is the image of the pencil? (Give the location of the images of the points $A$, $B$, and $C$ on the object, which are located at the eraser, point, and center of the pencil, respectively.) (b) What is the length of the image (that is, the distance between the images of points $A$ and $B$)? (c) Show the orientation of the image in a sketch.

Figure **P34.106**



**34.107** ••• **BIO** People with normal vision cannot focus their eyes underwater if they aren't wearing a face mask or goggles and there is water in contact with their eyes (see Discussion Question Q34.23). (a) Why not? (b) With the simplified model of the eye described in Exercise 34.52, what corrective lens (specified by focal length as measured in air) would be needed to enable a person underwater to focus an infinitely distant object? (Be careful—the focal length of a lens underwater is *not* the same as in air! See Problem 34.92. Assume that the corrective lens has a refractive index of 1.62 and that the lens is used in eyeglasses, not goggles, so there is water on both sides of the lens. Assume that the eyeglasses are 2.00 cm in front of the eye.)

## MCAT-STYLE PASSAGE PROBLEMS

**BIO Amphibian Vision.** The eyes of amphibians such as frogs have a much flatter cornea but a more strongly curved (almost spherical) lens than do the eyes of air-dwelling mammals. In mammalian eyes, the shape (and therefore the focal length) of the lens changes to enable the eye to focus at different distances. In amphibian eyes, the shape of the lens doesn't change. Amphibians focus on objects at different distances by using specialized muscles to move the lens closer to or farther from the retina, like the focusing mechanism of a camera. In air, most frogs are nearsighted; correcting the distance vision of a typical frog in air would require contact lenses with a power of about −6.0 D.

**34.108** A frog can see an insect clearly at a distance of 10 cm. At that point the effective distance from the lens to the retina is 8 mm. If the insect moves 5 cm farther from the frog, by how much and in which direction does the lens of the frog's eye have to move to keep the insect in focus? (a) 0.02 cm, toward the retina; (b) 0.02 cm, away from the retina; (c) 0.06 cm, toward the retina; (d) 0.06 cm, away from the retina.

**34.109** What is the farthest distance at which a typical "nearsighted" frog can see clearly in air? (a) 12 m; (b) 6.0 m; (c) 80 cm; (d) 17 cm.

**34.110** Given that frogs are nearsighted in air, which statement is most likely to be true about their vision in water? (a) They are even more nearsighted; because water has a higher index of refraction than air, a frog's ability to focus light increases in water. (b) They are less nearsighted, because the cornea is less effective at refracting light in water than in air. (c) Their vision is no different, because only structures that are internal to the eye can affect the eye's ability to focus. (d) The images projected on the retina are no longer inverted, because the eye in water functions as a diverging lens rather than a converging lens.

**34.111** To determine whether a frog can judge distance by means of the amount its lens must move to focus on an object, researchers covered one eye with an opaque material. An insect was placed in front of the frog, and the distance that the frog snapped its tongue out to catch the insect was measured with high-speed video. The experiment was repeated with a contact lens over the eye to determine whether the frog could correctly judge the distance under these conditions. If such an experiment is performed twice, once with a lens of power −9 D and once with a lens of power −15 D, in which case does the frog have to focus at a shorter distance, and why? (a) With the −9 D lens; because the lenses are diverging, the lens with the longer focal length creates an image that is closer to the frog. (b) With the −15 D lens; because the lenses are diverging, the lens with the shorter focal length creates an image that is closer to the frog. (c) With the −9 D lens; because the lenses are converging, the lens with the longer focal length creates a larger real image. (d) With the −15 D lens; because the lenses are converging, the lens with the shorter focal length creates a larger real image.

# ANSWERS

## Chapter Opening Question ?

**(ii)** A magnifying lens (simple magnifier) produces a virtual image with a large angular size that is infinitely far away, so you can see it in sharp focus with your eyes relaxed. (A surgeon would not appreciate having to strain her eyes while working.) The object should be at the focal point of the lens, so the object and lens are separated by one focal length. The distance from magnifier to eye is not crucial.

## Key Example VARIATION Problems

**VP34.4.1** (a) 14.5 cm (b) 29.0 cm (c) −30.0
**VP34.4.2** (a) $s' = -27.1$ cm, $m = +2.47$, virtual, erect, larger
(b) $s' = +45.9$ cm, $m = -1.48$, real, inverted, larger
(c) $s' = +27.9$ cm, $m = -0.507$, real, inverted, smaller
**VP34.4.3** (a) convex (b) 99.0 cm (c) +0.182, virtual, erect, smaller
**VP34.4.4** (a) $s' = -6.90$ cm, $m = +0.627$, virtual, erect, smaller
(b) $s' = -11.6$ cm, $m = +0.374$, virtual, erect, smaller
(c) $s' = -13.8$ cm, $m = +0.252$, virtual, erect, smaller
**VP34.8.1** (a) +39.0 cm (b) −39.0 cm
**VP34.8.2** (a) center (b) +68 cm (c) converging
**VP34.8.3** (a) edges (b) −68 cm (c) diverging
**VP34.8.4** (a) −15 cm (b) convex
**VP34.10.1** (a) −37.5 cm, same side (b) +2.50 (c) virtual, erect, larger
**VP34.10.2** (a) +16.8 cm, converging (b) −1.50 (c) real, inverted, larger
**VP34.10.3** (a) converging (b) +48.0 cm (c) +24.0 cm
**VP34.10.4** (a) +0.200 (b) −50.0 cm, same side (c) −62.5 cm, diverging

## Bridging Problem

(a) 29.9 cm to the left of the goblet
(b) 3.73 cm to the right of the goblet